MTech (CSE/AI) Course Structure

AY 2025-26 onwards

The revised MTech course structure for CSE/AI comprises the following courses, aligned with the specialization in the streams of AI/ML, Systems, and Theory.
MTech in AI is deemed to be in the AI-ML stream only and should follow the corresponding curriculum.

1. CS 610 (Algorithms) is a compulsory course for all streams (Offered in Odd semester).
2. CS 621 (Ethics of AI) is a compulsory course for all streams (Offered in Even semester).

Students in each of these streams take:

1. the core courses (in red below - 8 credits)
2. at least 4 credits in the respective baskets (in blue below)
3. at least 4 credits from any of the other two baskets

AI/ML	Systems	Theory
Core: ES 678: Machine Learning ES 667: Deep Learning	Core: CS 612: Computer Systems CS 616: Distributed Systems and Cloud Computing	Core: CS 619: CS Theory Toolkit CS 614: Advanced Algrithms
AI/ML Basket	Systems Basket	Theory Basket

AI/ML Basket

1. CS 613: NLP
2. ES 661: Probabilistic ML
3. ES 645: Optimization for ML
4. ES 659: Computer Graphics
5. ES 606: Computational neuroscience
6. CS 618: Theoretical Foundations of ML
7. ES 404: Networks and Complex Systems
8. ES 670 Matrix Methods for Signal Processing, Data Science, and Machine Learning
9. ES 666: Computer Vision
10. ES 612: Artificial Intelligence,
11. CS 618: Theoretical Foundations of ML
12. CS 329: Foundations of AI: Multiagent Systems

13. ES 634: Applied Multivariate Data Analysis
14. ES 615: Nature-inspired computing,

Systems Basket

1. CS 615: Advanced Networks
2. ES 417: 5G and Beyond
3. CS 434: Software Engineering and Testing
4. CS 435: Human-Computer Interaction
5. CS 431: Computer and Network Security
6. CS 691-XIV: Special Topics: Usable Privacy and Security

Theory Basket

1. CS 617: Complexity Theory
2. CS 607: Combinatorics in CS
3. ES 627: Linear Algebra and Computation
4. ES 643: Quantum computation and information
5. MA 607: Graph Theory

Additionally, according to the new MTech credit structure, the remaining credits for the degree requirement can be fulfilled by any of the above courses or the following options.

1. **Any** CS xxx tagged course (offline)
2. Any of the above courses that are mentioned in the above baskets.

For any < 600 level course, the usual constraint on the number of such credits permissible applies. The above list and buckets will be revised periodically based on the list of new courses offered by the CS faculty. Note that no online courses count here.